LINUX ESSENTIALS
CHAPTER 7
ARCHIVING AND COMPRESSION

Prepared by
Eng. Ibrahim Reyad
Email: ibrahimreyad1@gmail.com



Objective

In this lab, you will perform the following tasks:

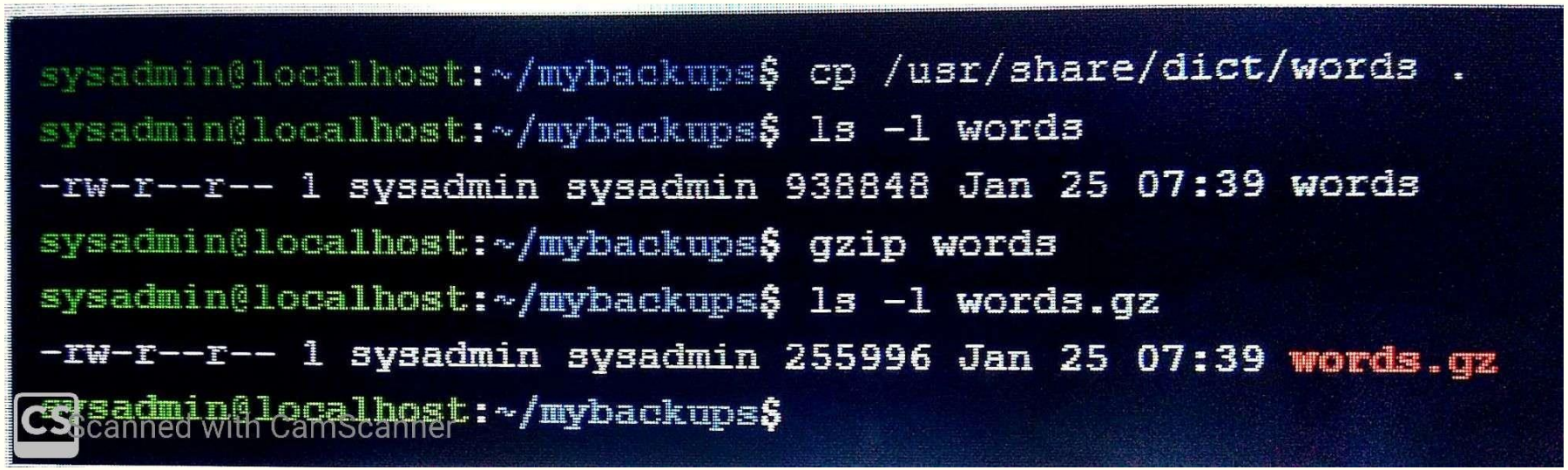
- Create archive files using tar with and without compression
- Compress and uncompress files into a gzip archive file
- Compress and uncompress files into a bzip2 archive file
- Use zip and unzip to compress and uncompress archive files

ARCHIVING COMMANDS

- In this task, we will use gzip, bzip2, tar and zip/unzip to archive and restore files. These commands are designed to either merge multiple files into a single file or compress a large file into a smaller one. In some cases the commands will perform both functions.

Compression Commands

- you will use gzip and gunzip to compress and uncompress a file
- Execute the following commands to compress a copy of the words file:



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Compression Commands

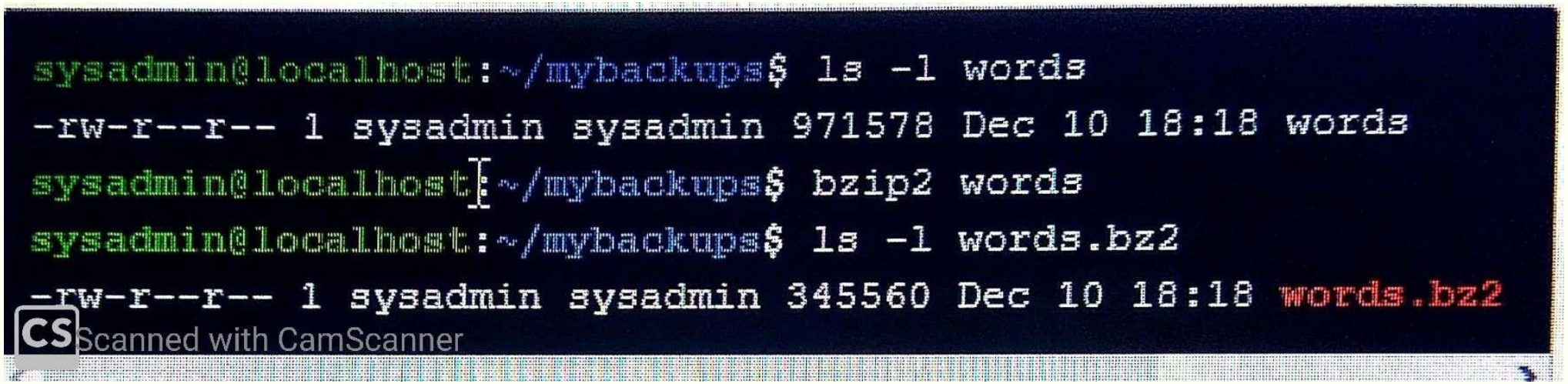
- Execute the following commands to uncompress the words.gz file:

```
sysadmin@localhost:~/mybackups$ ls -l words.gz
-rw-r--r-- 1 sysadmin sysadmin 259983 Dec 10 18:18 words.gz
sysadmin@localhost:~/mybackups$ gunzip words.gz
sysadmin@localhost:~/mybackups$ ls -l words
-rw-r--r-- 1 sysadmin sysadmin 971578 Dec 10 18:18 words
sysadmin@localhost:~/mybackups$
```

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Compression Commands

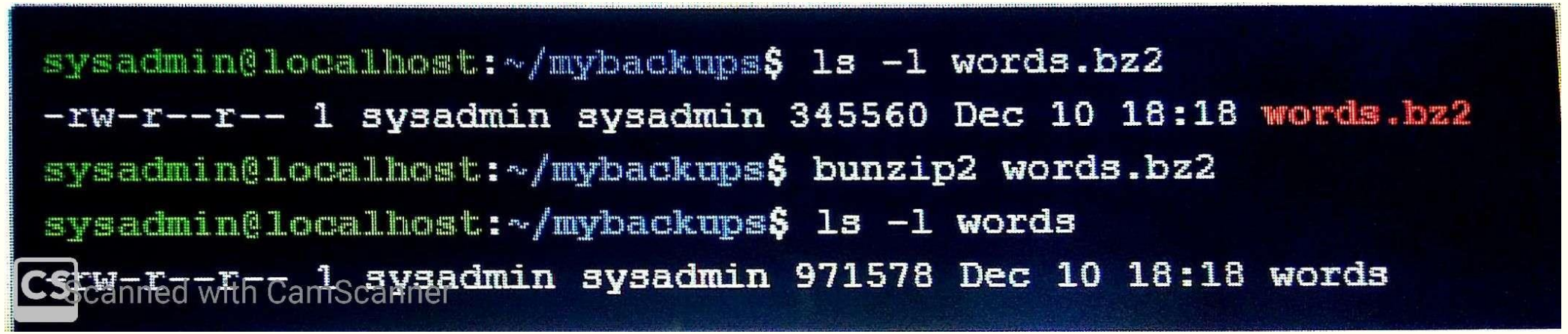
- Using bzip2 and bunzip2 to compress and uncompress a file is very similar to using gzip and gunzip.
- The compressed file is created with a .bz2extension. The extension is removed when uncompressed. Execute the following commands to compress a copy of the words file:



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Compression Commands

- Execute the following commands to uncompress the words.bz2 file:



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Archiving Commands

The task of archiving data is important for several reasons, including but not limited to the following:

- Large files may be difficult to transfer. Making these files smaller helps make transfer quicker.
- Transferring multiple files from one system to another can become tedious when there are many files. Merging them into a single file for transport makes this process easier.
- Files can quickly take up a lot of space, especially on smaller removable media like thumb drives. Archiving reduces this problem.

Archiving Commands

Tar has 3 modes:

- Create : make new archive.
- Extract : pull one or more files out of archive
- List : show the content of the archive without extracting

Archiving Commands

- Use the following tar command to create an archive of the /etc/udev directory.
- Save the backup in the ~/mybackups directory:

```
sysadmin@localhost:~$ cd
sysadmin@localhost:~$ mkdir mybackups
sysadmin@localhost:~$ tar -cvf mybackups/udev.tar /etc/udev
tar: Removing leading `/' from member names
/etc/udev/
/etc/udev/udev.conf
/etc/udev/hwdb.d/
/etc/udev/rules.d/
sysadmin@localhost:~$ ls mybackups
udev.tar
sysadmin@localhost:~$
```

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Archiving Commands

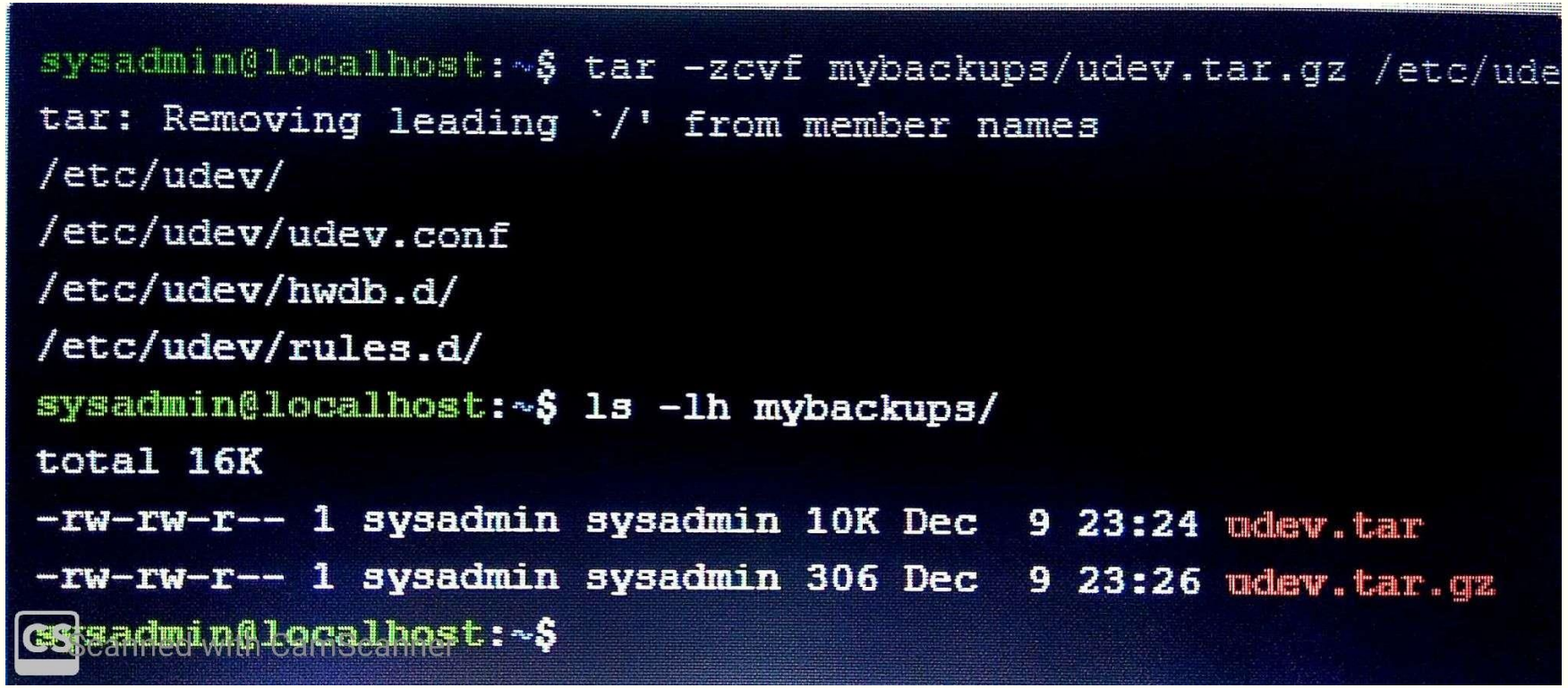
- Display the contents of a tar file by using the available options (t = list contents, v = verbose, f = filename):

```
sysadmin@localhost:~$ tar -tvf mybackups/udev.tar
drwxr-xr-x root/root         0 2018-07-19 06:51 etc/udev/
-rw-r--r-- root/root       153 2018-04-20 16:55 etc/udev/udev.
drwxr-xr-x root/root         0 2018-04-20 16:55 etc/udev/hwdb.
drwxr-xr-x root/root         0 2018-04-20 16:55 etc/udev/rules
sysadmin@localhost:~$
```

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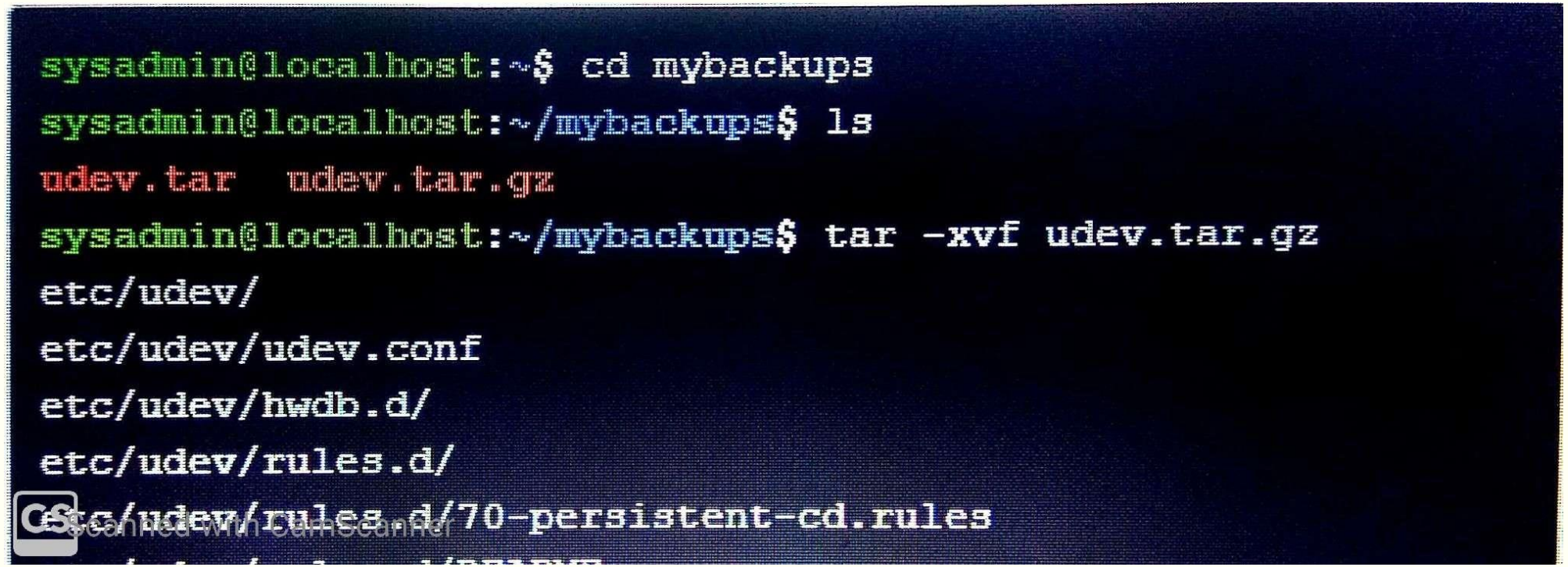
Archiving Commands

- To create a tar file that is compressed use -z option: The -z option makes use of the gzip utility to perform compression.



Archiving Commands

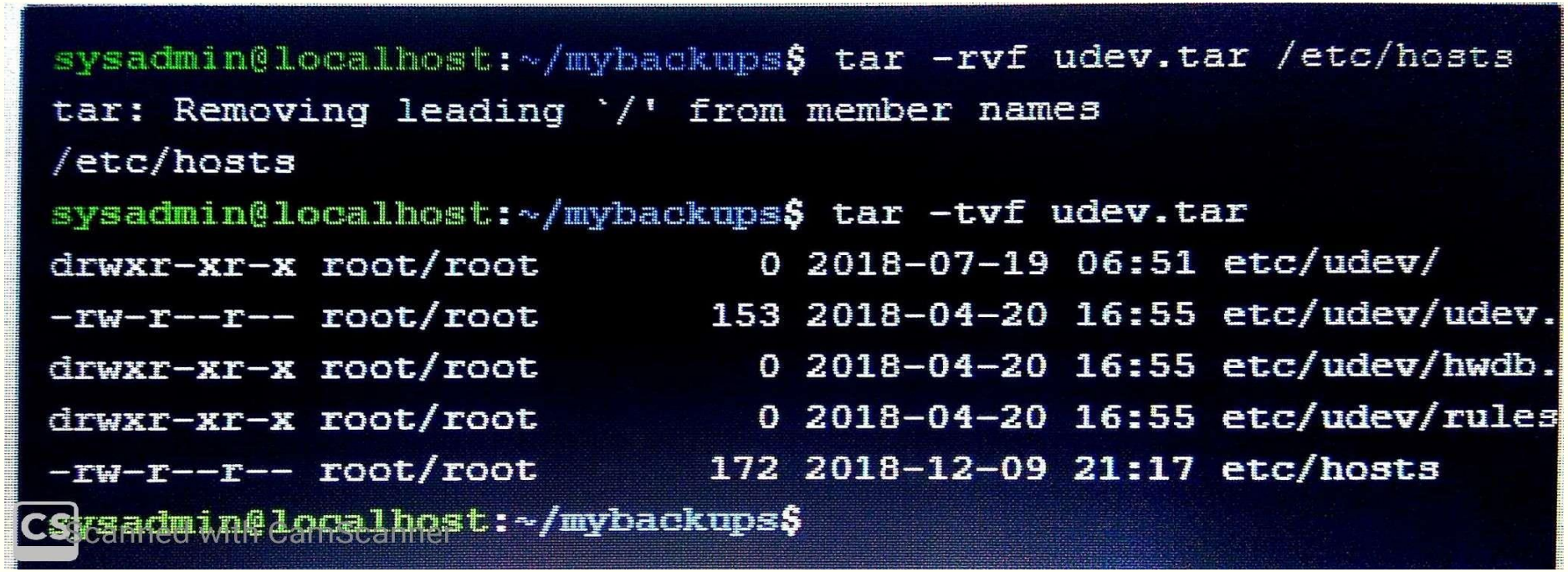
- Extract the contents of an archive. Data is restored to the current directory by default:



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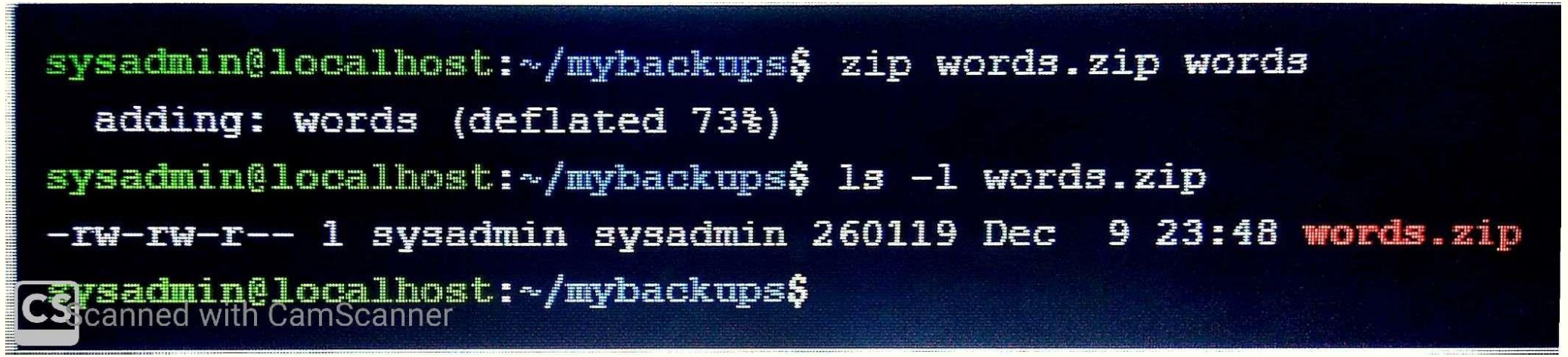
Archiving Commands

- To add a file to an existing archive, use the -r option to the tar command. Execute the following commands to perform this action and verify the existence of the new file in the tar archive:



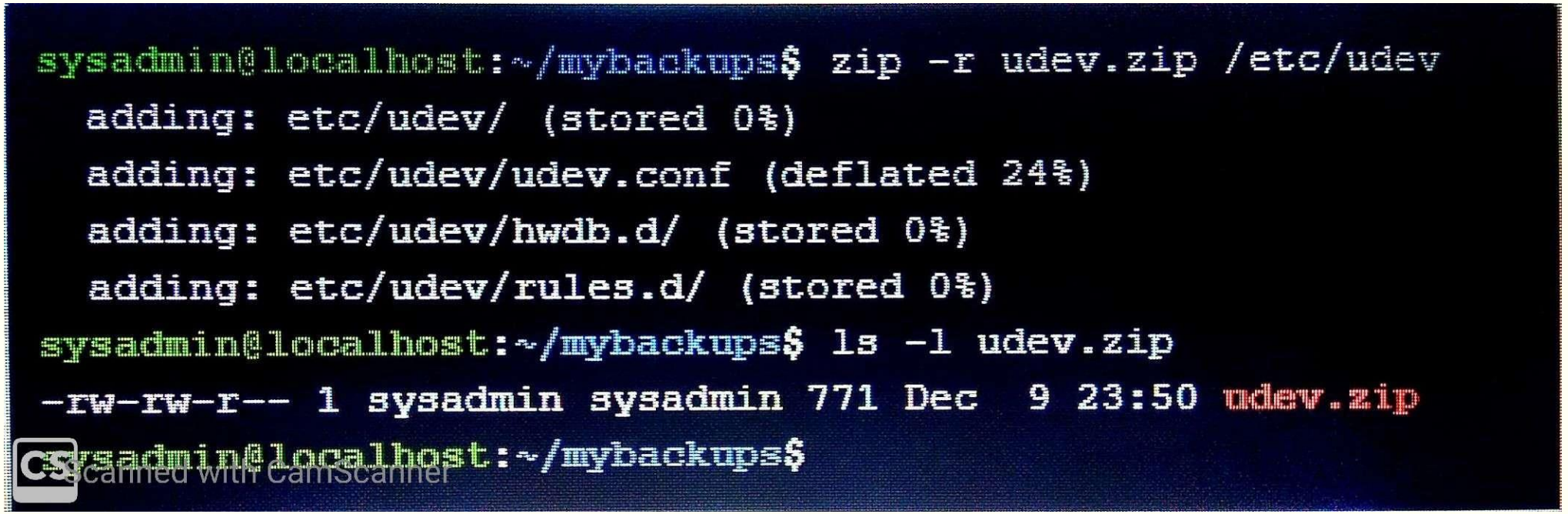
Compression Commands

- Use the zip command to compress the words file:



Compression Commands

- Compress the /etc/udev directory and its contents with zip compression:



Compression Commands

- To view the contents of a zip archive, use with the -l option with the unzip command:

```
sysadmin@localhost:~/mybackups$ unzip -l udev.zip
Archive:  udev.zip
  Length      Date    Time    Name
---------  ---------- -----   ----
        0  2018-07-19 06:51   etc/udev/
      153  2018-04-20 16:55   etc/udev/udev.conf
        0  2018-04-20 16:55   etc/udev/hwdb.d/
        0  2018-04-20 16:55   etc/udev/rules.d/
---------                     -------
      153                     4 files
sysadmin@localhost:~/mybackups$
```

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Compression Commands

- To extract the zip archive, use the unzip command without any options. In this example we first need to delete the files that were created in the earlier tar example:

```
sysadmin@localhost:~/mybackups$ rm -r etc
sysadmin@localhost:~/mybackups$ unzip udev.zip
Archive:  udev.zip
   creating: etc/udev/
  inflating: etc/udev/udev.conf
   creating: etc/udev/hwdb.d/
   creating: etc/udev/rules.d/
sysadmin@localhost:~/mybackups$
```

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